

Dynatrace Advanced Workshop — 2 Days

Audience: Vodafone Engineering & Operations Teams

Focus: Intermediate to Advanced (70% Hands-On)

Mode: Instructor-led, Lab-based Practical Workshop

Day 1 – Advanced Monitoring, Diagnostics & Troubleshooting

Module 1: Kickoff and Workshop Orientation

- Overview of the workshop goals and structure
- Quick recap of Dynatrace fundamentals and current Vodafone setup context
- Environment validation and access setup for all participants



Hands-On:

- Validate Dynatrace access (tenant, environments, roles)
 - Explore Smartscape topology for familiarization
 - Quick check: Identify monitored hosts, services, and applications
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Module 2: Deep Dive into Infrastructure & Application Monitoring

- Advanced host and process group analysis
- **Windows and Linux monitoring insights** (system metrics, process analytics, network diagnostics)
- Windows-specific monitoring (event logs, CPU queue length, service states, IIS metrics)
- Advanced web application monitoring and service flow visualization
- End-to-end dependency tracing and problem correlation using PurePath



Hands-On:

- Analyze Windows and Linux host performance using Smartscape & Processes view
 - Investigate IIS-based web application behavior
 - Perform end-to-end service trace analysis for a web app
 - Detect and analyze anomalies using Dynatrace's AI-driven detection
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Module 3: Database Monitoring and Query-Level Insights

- How Dynatrace detects and monitors database services
- Deep analysis of slow queries and connection bottlenecks
- Mapping database performance to application behavior
- Custom metrics and tagging for DB components



Hands-On:

- Identify and troubleshoot slow SQL queries
 - Correlate database latency with impacted service calls
 - Create custom DB metric tiles on a dashboard
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Module 4: Advanced Log Monitoring and Analysis

- Architecture of Dynatrace Log Monitoring (Grail-powered)
- Log ingestion, retention, and filtering best practices
- Creating DQL-based queries for log analytics
- Integrating logs into alerts and dashboards
- Handling Windows Event Logs and Syslog sources

Hands-On:

- Ingest and query application and Windows system logs
 - Build DQL queries to identify error patterns and anomalies
 - Visualize log data and link it with service traces
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Module 5: Real-World Troubleshooting Scenario

- Simulating end-to-end issue: app slowness or outage
- Using Dynatrace AI and full-stack views for Root Cause Analysis (RCA)
- Best practices for communicating findings and resolution paths

Hands-On:

- Work through a guided incident scenario
 - Use logs, traces, and metrics together for diagnosis
 - Document findings in a collaborative format
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Day 2 – Customization, Integration & Automation

Module 6: Building Advanced Dashboards & Using Davis AI

- Deep dive into the Data Explorer
- Leveraging Davis AI insights for dashboards
- Advanced visualization techniques and dynamic tiles
- Multi-team dashboards and sharing best practices

Hands-On:

- Create a performance-focused dashboard with AI insights
 - Add logs, metrics, and custom tiles for an end-to-end view
 - Use tags and filters to customize dashboard visibility
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Module 7: Azure Cloud Integration

- Integrating Dynatrace with **Microsoft Azure**
- Monitoring key Azure services:
 - **Azure Virtual Machines (Windows & Linux)**
 - **Azure SQL Database**
 - **Azure App Service**
 - **Azure Kubernetes Service (AKS)**
 - **Azure Functions**
- Azure Monitor metric ingestion and custom dimensions
- Cloud-native troubleshooting using Dynatrace

Hands-On:

- Connect Dynatrace with Azure subscription (using demo credentials or mock environment)
 - Visualize Azure resources and topology in Smartscape
 - Add Azure metrics and logs to an existing dashboard
 - Monitor performance of a Windows-based Azure VM
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Module 8: Automation & API-Driven Configuration

- Overview of Dynatrace APIs for automation and integration
- Using Configuration API for Infrastructure as Code (IaC)
- Automating alert and dashboard creation
- Integrating Dynatrace with CI/CD tools (Azure DevOps, Jenkins, GitLab, etc.)

Hands-On:

- Call Dynatrace APIs for metrics and problem detection
 - Automate dashboard creation with JSON templates
 - Build a CI/CD pipeline in Azure DevOps that triggers Dynatrace problem events
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Module 9: Best Practices & Governance

- Effective tagging strategies and management zones
- Role-based access control and governance
- Managing multiple environments and data privacy
- Performance optimization tips for OneAgent, ActiveGate, and Azure integration

Hands-On:

- Create a management zone and assign users
 - Implement tagging rule sets for environment organization
 - Troubleshoot OneAgent and ActiveGate connection issues on Azure-hosted VMs
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Module 10: Final Project & Knowledge Wrap-Up

- Group activity: build an integrated monitoring view for a simulated Azure-based application
- Present dashboards, alerts, and RCA workflows
- Review of workshop takeaways, best practices, and next steps

Hands-On (Capstone):

- Combine metrics, logs, and **Azure** resource data into one advanced dashboard
- Configure Davis AI alerts and validate problem notifications
- Present key insights and recommendations

Deliverables

- Workshop deck and reference notes
- Custom dashboard templates and DQL query samples
- API scripts for automation exercises
- Post-training guide and quick reference sheet